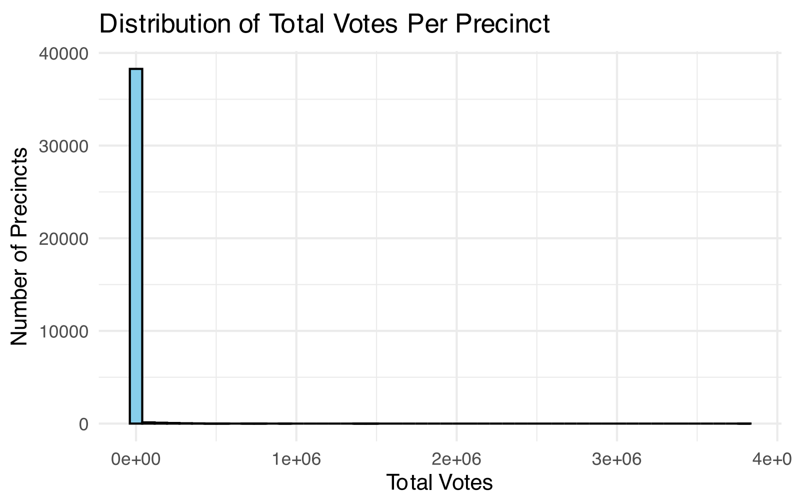
Exploratory Data Analysis

Question 1

How is the distribution of the total votes across all constituencies?

Answer 1

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
      0       1      91    2202     446 3793980 
```



The median was 91 votes, clearly indicating that the voting scale in most individual constituencies was relatively small. However, the mean (2,202 votes) was much higher than the median, and the maximum value reached an astonishing 3,793,980 votes, causing the distribution to exhibit a highly right-skewed pattern.

Question 2

Which county has the highest turnout rate for the general election?

Answer 2

```
[1] "Top 5 Counties by Total Voter Turnout"
```

```
# A tibble: 5 × 2
  fips  total_turnout
  <chr>         <dbl>
1 06037      26556986
2 06073      10521126
```

```
3 06059       9921779
4 06065       6713686
5 06071       5402838
```

Question 3

Which electoral district has the smallest percentage difference in the voting results for the US Senate race?

Answer 3

```
[1] "Closest US Senate Race Precinct"
```

```
# A tibble: 1 × 6
  fips  svprec_key  ussdem01 ussrep01 sen_total sen_pct_margin
  <chr> <chr>          <dbl>    <dbl>     <dbl>          <dbl>
1 06001 06001520810       80       80       160              0
```

The analysis of the US Senate election reveals a perfect tie in the 06001520810 district. The Democratic candidate in this district received 80 votes, and the Republican candidate also received 80 votes, resulting in a 0% difference in the vote rate. This outcome represents the smallest victory margin discovered in this exploration, which is the closest election. Although the total number of votes in this district is relatively small, only 160, it provides a vivid example at the micro level of how voters are perfectly divided, indicating that at this specific location, the preferences of voters have reached a complete balance, reflecting extreme competitiveness.